

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2025 INVITATIONAL A**

Biology

B01. D
B02. C
B03. D
B04. A
B05. E
B06. B
B07. A
B08. B
B09. C
B10. E
B11. C
B12. E
B13. E
B14. D
B15. B
B16. D
B17. A
B18. A
B19. B
B20. C

Chemistry

C01. E
C02. C
C03. B
C04. A
C05. E
C06. D
C07. B
C08. C
C09. A
C10. D
C11. A
C12. C
C13. D
C14. B
C15. D
C16. A
C17. E
C18. A
C19. B
C20. E

Physics

P01. C
P02. B
P03. E
P04. C
P05. A
P06. C
P07. B
P08. D
P09. D
P10. E
P11. D
P12. A
P13. B
P14. C
P15. B
P16. E
P17. E
P18. A
P19. C
P20. D

CHEMISTRY SOLUTIONS – UIL INVITATIONAL A 2025

- C01. (E) $2 \times 39.10 + 12.01 + 3 \times 16.00 = 138.21 \text{ g/mol}$
- C02. (C) $1 \text{ H}_3\text{PO}_4(\text{aq}) + 3 \text{ KOH}(\text{aq}) \rightarrow 1 \text{ K}_3\text{PO}_4(\text{aq}) + 3 \text{ H}_2\text{O}(\ell)$
- C03. (B) Electrons falling from higher energy levels to lower energy levels in an atom give off the difference in energy as light.
- C04. (A) In valence bond theory, s , p , and d orbitals in an isolated atom can combine to form an equal number of degenerate energy hybridized orbitals. The names of the hybridized orbitals are the isolated atomic orbitals that went into creating the hybridized orbital, so sp^3 hybridized orbitals are made up of one s and three p orbitals for a total of four equal-energy sp^3 orbitals, allowing the atom to form four equal-energy single bonds like the carbon atom in methane.
- C05. (E) $PV = nRT$ so $T = PV/nR$. $N = 1$, $V = 44.8 \text{ L}$, $P = 1 \text{ atm}$. $R = 0.08206 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
 $T = (1)(44.8)/(1)(0.08206) = 545.94 \text{ K} = 273 \text{ }^\circ\text{C}$. Or since you are doubling the volume, you need to double the temperature, from 273 K (STP) to $546 \text{ K} = 273 \text{ }^\circ\text{C}$.
- C06. (C) The liquid in the thermometer “rises” because it expands by increasing the amount of space between the particles. At higher temperatures the thermometer will have the same number of particles in it and they will still be the same size, but they will be moving faster and will be spaced farther apart.
- C07. (B) The new concentration is one half of the original, or 0.05 M . $-\log(0.05) = 1.3$
- C08. (C) The molar mass of propanol is 60.11 , so 100.0 grams is 1.664 moles .
 The *change in heat* is $1.664 \text{ moles} \times -2021 \text{ kJ/mol} = -3362 \text{ kJ}$, so the amount of heat *given off* is 3362 kJ .
- C09. (A) $-m_{\text{Al}}c_{\text{Al}}\Delta T_{\text{Al}} = m_{\text{water}}c_{\text{water}}\Delta T_{\text{water}}$
 $c_{\text{Al}} = 0.897 \text{ J/g}^\circ\text{C}$
 $\Delta T_{\text{Al}} = (25.8 - 88.0) = -62.2 \text{ }^\circ\text{C}$
 $m_{\text{water}} = 125 \text{ g}$
 $c_{\text{water}} = 4.184 \text{ J/g}^\circ\text{C}$
 $\Delta T_{\text{water}} = (25.8 - 25.0) = 0.8 \text{ }^\circ\text{C}$
- $$m_{\text{Al}} = \frac{-m_{\text{w}}c_{\text{w}}\Delta T_{\text{w}}}{c_{\text{Al}}\Delta T_{\text{Al}}} = \frac{-(125)(4.184)(0.8)}{(0.897)(-62.2)} = 7.5 \text{ g}$$
- (ΔT_{water} = limits our final answer to only one significant digits.)
- C10. (D) The equilibrium constant K is determined by the ratio of the forward and backward reaction rates.
- C11. (A) AgCl is insoluble in water. None of the other combinations result in a product that is insoluble in water.
- C12. (C) Ordinary lab conditions are 1 atm pressure and about 25°C , which is the liquid region on the graph.
- C13. (D)

- C14. (B) Complete combustion reactions of hydrocarbons (as well as oxygen-containing organic compounds) always produce CO_2 and H_2O as the products. Incomplete combustion (which is the norm in real life fires) produces all kinds of partial combustion products including carbon monoxide.
- C15. (D) The Formal charge on an atom in a compound assumes all electrons are shared equally and oxidation numbers treat all bonds as if they are ionic. Although oxidation numbers are the same as the charge on a monatomic ion, the oxidation numbers on atoms in a polyatomic ion are not the same as the charge on the ion (but they all add up to the total charge).
- C16. (A) $M_1V_1 = M_2V_2$. $M_1 = 5.00$, $V_1 = 75.0$, $M_2 = 4.00$, $V_2 = (5.00)(75.0)/4.00 = 93.75$ mL. that is the *total* volume of the 4.00 M solution, and you start with 75.0 mL of stock, so you need to add $93.75 - 75.0 = 18.75$ mL.
- C17. (E) When naming covalent compounds you do not put a *mono-* prefix on the first element in the compound name. For example, it's carbon dioxide and carbon monoxide, not monocarbon dioxide and monocarbon monoxide. The *mono-* prefix is used only on the second element in the compound name in a binary covalent compound. PO is an unstable compound called *phosphorus monoxide*.
- C18. (A) $225 \text{ g} / 26.98 \text{ g/mol} = 8.3395 \text{ mol Al} = 8.3395 \text{ mol AlF}_3$.
 $8.3395 \text{ mol AlF}_3 \times 83.98 \text{ g/mol} = 700.35 \text{ grams AlF}_3$.
- C19. (B) $PV=nRT$, so $n = PV/RT$. $T = 79+273=352 \text{ K}$. $n = (1)(16)/(0.08206 \times 352) = 0.5539$ moles of $\text{CO}_2 \times 6.022 \times 10^{23} = 3.336 \times 10^{23}$ molecules. There are three atoms per CO_2 molecule, so number of atoms = $3.336 \times 10^{23} \text{ molecules} \times 3 \text{ atoms/molecule} = 1.00 \times 10^{24} \text{ atoms}$.
- C20. (E) B and C both have the effect of increasing the surface area of the solid, allowing for more interactions with the aqueous reactant. BTW, there is one more way to speed up a chemical reaction that is not listed here...

PHYSICS SOLUTIONS – UIL INVITATIONAL A 2025

- P01. (C) page 23: “Observers moving at constant speed relative to each other define a special group of inertial frames, so called because they are frames of reference in which Newton’s principle of inertia (the first law) holds true.”
- P02. (B) pages 34-35: “While Maxwell’s Equations may look intimidating in mathematical notation, they’re actually very simple to translate into English.... The four equations translate into four rules:
1. The strength of an electric field depends on the amount of charge in the vicinity.
 2. A magnet will always have both north and south poles.
 3. An electric field is created by a changing magnetic field.
 4. A magnetic field is created by either a current or a changing electric field.”
- P03. (E) page 40: “The most famous experiment to look for the effects of motion on the speed of light is known to physicists as the Michaelson-Morley experiment; it was carried out by American physicist Albert Michaelson with the assistance of Edward Morley, starting in the 1880s.”
- P04. (C) Venus’s atmosphere is composed of about 95% carbon dioxide. It has some nitrogen (3%) and trace amounts of sulfur dioxide, carbon monoxide, water vapor and argon. The high level of carbon dioxide has caused a runaway greenhouse effect, making Venus the hottest planet in the solar system.
- P05. (A) First, we should convert the given quantities to metric units. One meter equals 3.28 feet. This means that 1500 feet is $d = \frac{1500ft}{3.28ft/m} = 457m$. For time, we get $t = (2.8min) * (60 \frac{sec}{min}) = 168 s$. To find velocity, we divide distance by time, which gives: $v = \frac{d}{t} = \frac{455m}{168 s} = 2.7 m/s$.
- P06. (C) We do not know how long the acorn was in the air, but we do know the distance it traveled through the air. So, we’ll use the equation that does not involve time: $v_f^2 = v_i^2 + 2a\Delta y$. Plugging in the given values, we get $v_f^2 = (8.60)^2 + 2(9.80)(4.40) = 160.2 \rightarrow v_f = 12.7m/s$.
- P07. (B) We begin with a force diagram for the framed picture. There are three forces: gravity (mg , directed downward), tension on the left (T_L , directed up and right), and tension on the right (T_R , directed up and left). Since both tensions act at an angle, we must consider their components. In the x-direction, we have $T_L \cos \theta$ on the left, and $-T_R \cos \theta$ on the right. In the y-direction, we get $T_L \sin \theta$ on the left and $T_R \sin \theta$ on the right. The picture is in static equilibrium, so all of the forces in each direction must sum to zero. In the x-direction, we get $\sum F_x = T_L \cos \theta - T_R \cos \theta = 0$. This leads to $T_L \cos(24.0) = T_R \cos(24.0) \rightarrow T_L = T_R$. We could already surmise this based on the symmetry of the problem. Let’s call the tensions on each side T since they are the same $T = T_L = T_R$. For the y-direction, we get $\sum F_y = T_L \sin \theta + T_R \sin \theta - mg = 0$. Renaming the tensions, and putting in the numbers, we get $T \sin(24.0) + T \sin(24.0) = (1.20)(9.80)$. This gives $2T \sin(24.0) = 11.76 \rightarrow 0.8135T = 11.76 \rightarrow T = 14.4N$.
- P08. (D) Since we are given the initial kinetic energy, it makes sense to use conservation of energy to solve the problem. As the acorn rises, its kinetic energy is converted into gravitational potential energy. At its maximum height, all of the kinetic energy has become gravitational potential energy: $KE_i = GPE_f = 0.180J$. Putting in the equation for gravitational potential energy, we obtain the maximum height of the acorn:
 $GPE_f = mgh_f = 0.180 \rightarrow (2.50 \times 10^{-3})(9.80)h_f = 0.180$. This gives $h_f = 7.35m$.

- P09. (D) First, we need to convert the final angular velocity into the proper units:
 $\omega_f = 150 \frac{\text{rev}}{\text{min}} * \frac{2\pi \text{ rad}}{1 \text{ rev}} * \frac{1 \text{ min}}{60 \text{ sec}} = 15.71 \text{ rad/s}$. Now we use the angular kinematic equation:
 $\omega_f = \omega_i + \alpha t \rightarrow 15.71 = 0 + (30.0)t$ This leads to $t = 0.524 \text{ sec}$. Note: The radius of the disk is given but isn't needed.
- P10. (E) The period of a mass-spring system is given by the equation $T = 2\pi \sqrt{\frac{m}{k}}$. Rearranging to solve for the spring constant gives $k = \left(\frac{2\pi}{T}\right)^2 m$. Converting the mass into kilograms, and plugging in the known values, results in a spring constant of $k = \left(\frac{2\pi}{0.855}\right)^2 (0.300) = 16.2 \text{ N/m}$.
- P11. (D) The equation that describes thermal expansion is $\Delta L = L_0 \alpha \Delta T$. First, we'll find the change in length from cold to hot: $\Delta L = 3.324 - 3.311 = 0.013 \text{ m}$. Using the cold length as the original length, L_0 , we get: $0.013 = (3.311)\alpha(42 - (-15)) \rightarrow 0.013 = (3.311)(57)\alpha$. So, the coefficient of thermal expansion is $0.013 = 188.7\alpha \rightarrow \alpha = 6.9 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.
- P12. (A) The two middle resistors, the 410Ω and the 530Ω , are in series, so we'll start by combining them. This gives a middle branch resistance of $R_M = 410 + 530 = 940\Omega$. This middle resistance is in parallel with the upper resistor (330Ω). Branches in parallel will all have the same voltage across them, which means that the voltage across the 330Ω resistor is $V_U = 24.0\text{V}$, and the voltage across the middle branch is also $V_M = 24.0\text{V}$. This also means that we don't need to combine the last two resistances. Now that we have the middle branch voltage, we use Ohm's law to find the middle branch current: $V_M = I_M R_M \rightarrow 24 = I_M(940)$. This gives a middle branch current of $I_M = 0.0255 \text{ A}$. For resistors in series the currents flowing through each resistor will be the same as the total current in the branch, so the current through each of the middle branch resistors is that same value. That is, $I_{530} = I_{410} = I_M = 0.0255 \text{ A} = 25.5 \text{ mA}$.
- P13. (B) The equation for the electric potential produced by a point charge is $V = \frac{kQ}{r}$. Here r is the distance from the point charge to the observer's location (point P). In this problem, we have $r = 30.0 \text{ cm} - 5.00 \text{ cm} = 25.0 \text{ cm}$. Thus, the electric potential at the point P is
 $V = \frac{(8.99 \times 10^9)(33.0 \times 10^{-9})}{0.250} = 1190 \text{ V}$.
- P14. (C) A charged particle in a magnetic field will trace out a circle with a radius given by $r = \frac{mv}{qB}$. From the given values, we first find the radius: $r = \frac{26.0}{2} = 13.0 \text{ cm} = 0.130 \text{ m}$, and we convert the magnetic field into the proper units: $B = 450 \text{ G} = 0.0450 \text{ T}$. Now, using the mass and charge of a proton, and plugging in the given values we obtain $r = 0.130 = \frac{(1.67265 \times 10^{-27})v}{(1.602 \times 10^{-19})(0.0450)} \rightarrow$
 $0.130 = (2.32 \times 10^{-7})v$. This gives a velocity for the proton of $v = 5.60 \times 10^5 \text{ m/s}$.
- P15. (B) We'll use $\Phi_B = BA \cos \theta$ to find the initial magnetic flux passing through the loop of wire. Here the angle θ is measured from the normal line of the area of the loop (that is from a line that is perpendicular to the face of the loop). Since the magnetic field is perpendicular to the face of the loop, the magnetic field and the normal line are in the same direction. Thus, $\theta = 0^\circ$. We also need the area of the loop. The loop is rectangular, so the area is found using $A = lw = (0.300)(0.150) = 0.045 \text{ m}^2$. Converting the field strength into proper units, we find that the initial magnetic flux is $\Phi_{Bi} = B_i A \cos \theta = (0.0220 \text{ T})(0.045 \text{ m}^2) \cos 0^\circ = 9.90 \times 10^{-4} \text{ Tm}^2$. The final magnetic flux is zero since $B_f = 0 \rightarrow \Phi_{Bf} = 0$. Now we use Faraday's Law to determine the voltage (EMF) induced in the loop by the changing magnetic flux:
 $\mathcal{E} = -\frac{\Delta \Phi_B}{\Delta t} = -\frac{0 - 9.90 \times 10^{-4} \text{ Tm}^2}{62.0 \times 10^{-6} \text{ s}} = 15.97 \text{ V}$. Finally, we use Ohm's Law to find the current induced in the loop: $\mathcal{E} = IR \rightarrow 15.97 \text{ V} = I(1.20\Omega) \rightarrow I = 13.3 \text{ A}$.

- P16. (E) The focal length of the mirror is $f = \frac{R}{2} = \frac{200.0}{2} = 100.0\text{cm}$. Since the mirror is concave, the focal length is positive. Now, working entirely in centimeters, we find the location of the image of the moth: $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \rightarrow \frac{1}{56.0\text{cm}} + \frac{1}{q} = \frac{1}{100.0\text{cm}} \rightarrow q = -127.3\text{cm}$. Since the image location is negative, we know that the image is virtual. The magnification of the image is given by $M = -\frac{q}{p} = -\frac{-127.3\text{cm}}{56.0\text{cm}} = 2.27$. The magnification is positive, so the image is upright. The size of the wingspan of the image of the moth is then: $w' = Mw = (2.27)(5.00\text{cm}) = 11.4\text{cm}$.
- P17. (E) First, we need to find the energy (in eV) of the photons of light. To do this, we use $E_\gamma = \frac{1240\text{ eVnm}}{\lambda} = \frac{1240\text{ eVnm}}{455\text{ nm}} = 2.725\text{eV}$. Then we use the photoelectric effect equation to find the energy imparted to each electron as kinetic energy: $KE = E_\gamma - \phi = 2.725 - 2.28 = 0.445\text{eV}$. This energy is low enough that the electrons will not move at relativistic speeds. Thus, we can use the classical equation for the kinetic energy: $KE = \frac{1}{2}mv^2$. Converting the kinetic energy into Joules and putting in the mass of an electron gives: $KE = 0.445\text{eV}(1.602 \times 10^{-19}) = 7.13 \times 10^{-20}\text{J} = (0.5)(9.11 \times 10^{-31}\text{kg})v^2 \rightarrow v^2 = 1.57 \times 10^{11}$. This gives an emitted electron velocity of $v = 3.96 \times 10^5\text{ m/s}$.
- P18. (A) This radioactive decay chain involves the emission of three alpha particles, two negative beta particles, and two gamma rays. Alpha particles carry away four units of mass and two units of positive charge. Negative beta particles carry no units of mass but take away one unit of negative charge. Gamma rays carry neither mass nor charge (only energy). Based on this, the mass number of the daughter isotope will be:
 $M = 285 - 3\alpha - 2\beta^- - 2\gamma = 285 - 3(4) - 2(0) - 2(0) = 285 - 12 = 273$. And the charge (proton number) of the daughter isotope will be:
 $Z_d = 112 - 3\alpha - 2\beta^- - 2\gamma = 112 - 3(2) - 2(-1) - 2(0) = 112 - 6 + 2 = 108$. Thus, the daughter isotope will have a mass of 273 and a proton number of 108, which means that it is ${}_{108}^{273}\text{Hs}$.
- P19. (C) On a plot of velocity versus time, the acceleration is equal to the slope of the line. To find the slope, I'll choose two points on the line that aren't data points – specifically, I'll use (1.3s, 30.0m/s) and (7.0s, 10.0m/s). The slope (and the acceleration) is then:
 $\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10.0 - 30.0}{7.0 - 1.3} = \frac{-20.0\text{m/s}}{5.7\text{s}} = -3.5\text{ m/s}^2 = \text{acceleration}$.
- P20. (D) The important equation for this problem is Ohm's Law: $V = IR$. These data do not trace out a straight line, so we will use a single point on the curve to estimate the voltage. I'll choose the point (33.0Ω, 2.0A). Plugging these values into Ohm's Law gives $V = (2.0\text{A})(33.0\Omega) = 66\text{ V} \approx 65\text{ V}$.